



Recent Developments in Symbolic and Distributional Data Analysis

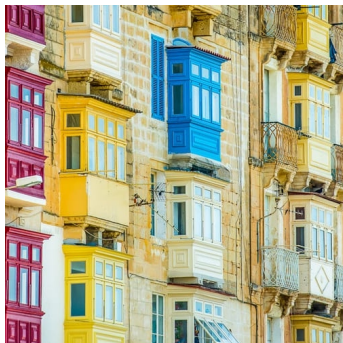
Symbolic networks

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5th ISI Regional Statistics Conference 2026
3-5 June 2026 in Valletta, Malta

- 1 Introduction
- 2 Networks and semirings
- 3 Activity
- 4 Examples
- 5 Conclusions
- 6 References



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Last version of slides (May 31, 2026, 23 : 56): [VBmalta26.pdf](#)

relational event: entity1, entity2, (type), start, (duration), value(s)

- Bike-sharing networks [5].
- Political event datasets KEDS.
- ENRON e-mails.
- World trade datasets such as BACI/CEPII [14].
- Analysis of temporal bibliographic data [3].

All these data can be described as **weighted non-simple** (a pair of nodes can be linked by multiple links) **networks**.

Symbolic Data Analysis (SDA) is an extension of traditional statistics and data mining that models complex, aggregated data. Instead of single numerical or categorical values, the input data – known as symbolic objects – contains internal variation and structure.

A **network** $\mathbf{N} = (V, L, P, W)$ consists of a **graph** $\mathbf{G} = (V, L)$, where

- V is the set of **nodes**
- $L = E \cup A$, $E \cap A = \emptyset$ is the set of **links** (A is the set of **arcs** – directed links, E is the set of **edges** – undirected links);
 $n = |V|$, $m = |L|$.
- P are node value functions / **properties**: $p : V \rightarrow R_p$; and
- W are link value functions / **weights**: $w : L \rightarrow R_w$.

Sometimes, implicit additional information/data about values is provided in the specifications of properties: (a) algebraic structures, and (b) properties of values.

A network $\mathbf{N}$ is **symbolic** if at least one of its value functions is symbolic.

Computing with link weights in networks

Symbolic
networks

V. Batagelj

Introduction

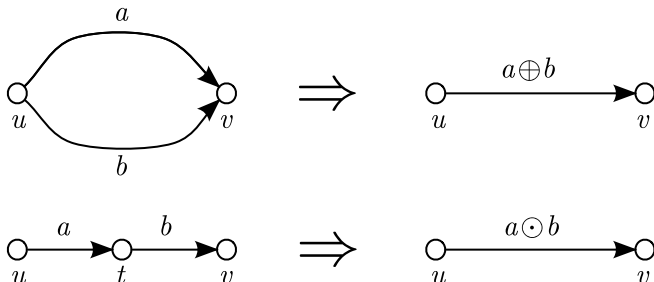
Networks and
semirings

Activity

Examples

Conclusions

References



A **semiring** is a "natural" algebraic structure to formalize computations with link weights in networks. They allow us to extend the weights from links to nodes, walks (paths), and sets of walks.

Let $\mathbb{A}$ be a set and a, b, c elements from $\mathbb{A}$. A *semiring* [10, 1, 15] is an algebraic structure $(\mathbb{A}, \oplus, \odot, 0, 1)$ with two binary operations (addition $\oplus$ and multiplication $\odot$) where:

$(\mathbb{A}, \oplus, 0)$ is an *abelian monoid* with the neutral element 0 (zero):

$$a \oplus b = b \oplus a \quad - \text{commutativity}$$

$$(a \oplus b) \oplus c = a \oplus (b \oplus c) \quad - \text{associativity}$$

$$a \oplus 0 = a \quad - \text{existence of zero}$$

$(\mathbb{A}, \odot, 1)$ is a *monoid* with the neutral element 1 (unit):

$$(a \odot b) \odot c = a \odot (b \odot c) \quad - \text{associativity}$$

$$a \odot 1 = 1 \odot a = a \quad - \text{existence of a unit}$$

Multiplication $\odot$ *distributes* over addition $\oplus$:

$$a \odot (b \oplus c) = a \odot b \oplus a \odot c \quad (b \oplus c) \odot a = b \odot a \oplus c \odot a$$

In formulas, we assume the precedence of multiplication over addition.

We can extend the weight w to walks and sets of walks on the network $\mathbf{N}$ by the following rules:

- Let $\sigma_v = (v)$ be a null walk in the node $v \in V$; then $w(\sigma_v) = 1$.
- Let $\sigma = (u_0, u_1, u_2, \dots, u_{p-1}, u_p)$ be a walk of length $p \geq 1$ on $\mathbf{N}$; then

$$w(\sigma) = \bigodot_{t=1}^p w(u_{t-1}, u_t).$$

- For the empty set of walks $\emptyset$, it holds $w(\emptyset) = 0$.
- Let $S = \{\sigma_1, \sigma_2, \dots\}$ be a set of walks in $\mathbf{N}$; then

$$w(S) = \bigoplus_{\sigma \in S} w(\sigma).$$

- 1 Combinatorial: $(\mathbb{N}, +, \cdot, 0, 1)$ or $(\mathbb{R}_0^+, +, \cdot, 0, 1)$
- 2 Reachability: $(\{0, 1\}, \vee, \wedge, 0, 1)$
- 3 Shortest paths: $(\mathbb{R}_0^+, \min, +, \infty, 0)$
- 4 MaxMin (capacity): $(\mathbb{R}_0^+, \max, \min, 0, \infty)$
- 5 Pathfinder [18]: $(\overline{\mathbb{R}}_0^+, \min, \boxplus, \infty, 0)$
where $a \boxplus b = \sqrt[r]{a^r + b^r}$ (Minkowski)
- 6 Regular languages [20] – regular algebra (union, concatenation, and iteration of sets of strings over an alphabet A).

- 7 Let $v : A \rightarrow B$ be a property/variable on the set of units A and the set of bins $\mathbf{B} = \{B_1, B_2, \dots, B_k\}$ be a partition of the set B . For a subset $C \subseteq A$, a **histogram** $h_C : \mathbf{B} \rightarrow \mathbb{N}$ is defined as $h_C(i) = h(B_i) = |\{x \in C : v(x) \in B_i\}|$

For histograms h and g on B , we will use the **addition**

$$h \oplus g : (h \oplus g)(i) = h(i) + g(i)$$

Their **multiplication** can be defined in different ways. We will use

$$h \odot g : (h \odot g)(i) = h(i) \cdot g(i)$$

Another option, assuming $(\mathbf{B}, \circ)$ is a semigroup, is the convolution $h * g$ [12, 9]

$$(h * g)(i) = \sum_{p \circ q = i} h(p) \cdot g(q)$$

Simplification of weighted networks

Symbolic
networks

V. Batagelj

Introduction

Networks and
semirings

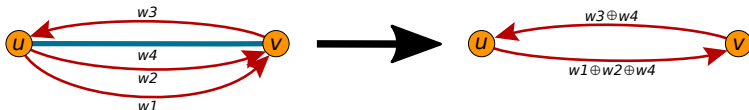
Activity

Examples

Conclusions

References

For a subset $S \subseteq L$ of links, we define its *total* weight as $\Sigma(S) = \bigoplus_{\ell \in S} w(\ell)$.



A *simplification* of a weighted network $\mathbf{N} = (V, L, w)$ over an abelian monoid $(\mathbb{A}, \oplus, 0)$ is its transformation into a *simple* directed weighted network $\hat{\mathbf{N}} = (V, A, \hat{w})$ such that for $L(u, v) = \{\ell \in L : \text{init}(\ell) = u \wedge \text{term}(\ell) = v\}$ it holds $L(u, v) \neq \emptyset \Rightarrow (u, v) \in A \wedge \hat{w}(u, v) = \Sigma(L(u, v))$.

For undirected weighted networks, a special undirected simplification can be defined similarly.

$$\text{loops}(v) = \{\ell \in L : \text{init}(\ell) = \text{term}(\ell) = v\}$$

$$\text{inStar}(v) = \{\ell \in L : \text{term}(\ell) = v \wedge \text{init}(\ell) \neq v\}$$

$$\text{outStar}(v) = \{\ell \in L : \text{init}(\ell) = v \wedge \text{term}(\ell) \neq v\}$$

Using the link weight w of the network $\mathbf{N}$, we can compute some interesting node properties such as *in-sum* (weighted indegree):

$$\text{inS}(\mathbf{N}, v) = \Sigma(\text{inStar}(v))$$

out-sum (weighted outdegree):

$$\text{outS}(\mathbf{N}, v) = \Sigma(\text{outStar}(v))$$

and *loop-sum* (weight of loops):

$$\text{loopS}(\mathbf{N}, v) = \Sigma(\text{loops}(v))$$

of a node v . They are describing the *activity* of the node v .

For a weighted network $\mathbf{N} = (V, L, w)$ and a subset of nodes $C \subseteq V$ we can define the **boundary links** of C :

$$\text{inBd}(C) = \{\ell \in L : \text{init}(\ell) \notin C \wedge \text{term}(\ell) \in C\}$$

$$\text{outBd}(C) = \{\ell \in L : \text{init}(\ell) \in C \wedge \text{term}(\ell) \notin C\}$$

$$\text{Bd}(C) = \text{inBd}(C) \cup \text{outBd}(C)$$

the **interior** of C : $\text{Int}(C) = \{\ell \in L : \text{init}(\ell) \in C \wedge \text{term}(\ell) \in C\}$

$$L(C, D) = \{\ell \in L : \text{init}(\ell) \in C \wedge \text{term}(\ell) \in D\}$$

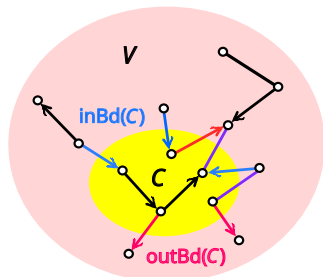
and the corresponding **activities**:

$$\text{inPut}(C) = \Sigma(\text{inBd}(C))$$

$$\text{outPut}(C) = \Sigma(\text{outBd}(C))$$

$$\text{Act}(C) = \Sigma(\text{Int}(C))$$

$$\text{Act}(C, D) = \Sigma(L(C, D))$$





Bike-sharing data analysis – Citi bike

Clustering of flows / 7 clusters [5, 2]

Symbolic
networks

V. Batagelj

Introduction

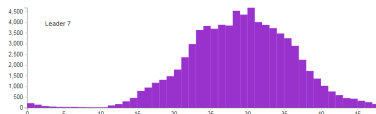
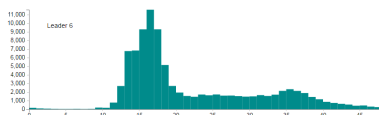
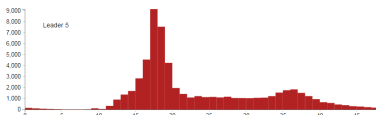
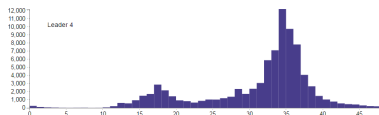
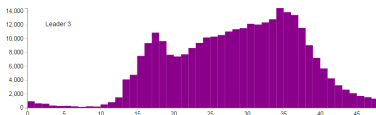
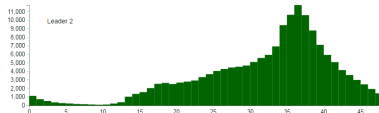
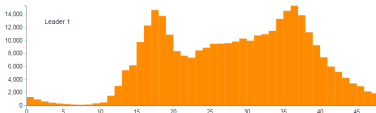
Networks and
semirings

Activity

Examples

Conclusions

References



NY Citi Bike one-year data from October 2015 to September 2016: 13266296 trips, 678 stations. ▶

V. Batagelj

Symbolic networks

Let the binary **affiliation** matrix $\mathbf{A} = [a_{ep}]$ describe a two-mode network on the set of events E and the set of participants P :

$$a_{ep} = \begin{cases} 1 & p \text{ participated at the event } e \\ 0 & \text{otherwise} \end{cases}$$

The function $d : E \rightarrow \mathcal{T}$ assigns to each event e the date $d(e)$ when it happened. Assume $\mathcal{T} = [first, last] \subset \mathbb{N}$. Using these data we can construct two temporal affiliation matrices [3]:

- **instantaneous** $\mathbf{Ai} = [ai_{ep}]$, where ai_{ep} is a vector on $\mathcal{T}$ with initial value $\mathbf{0}$. If $a_{ep} = 1$ then we set $ai_{ep}[d(e)] = 1$.
- **cumulative** $\mathbf{Ac} = [ac_{ep}]$, where ac_{ep} is initially $\mathbf{0}$. If $a_{ep} = 1$ then for each $i \in [d(e), last]$ we set $ac_{ep}[i] = 1$.

Temporal bibliographic networks

Temporal citations between journals

Symbolic
networks

V. Batagelj

Introduction

Networks and
semirings

Activity

Examples

Conclusions

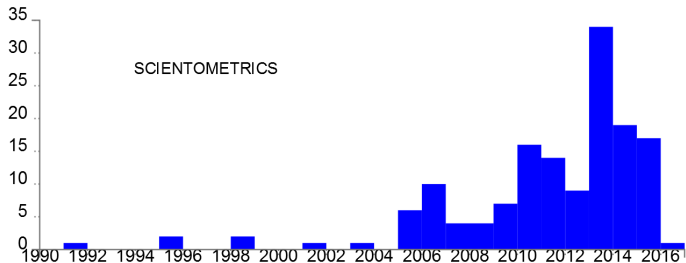
References

Networks are from the collection of bibliographic networks on peer review from WoS till 2017.

The derived network describing citations between journals is obtained as

$$\mathbf{JCJ} = \mathbf{WJi}^T \cdot \mathbf{CiI} \cdot \mathbf{WJc}$$

Note that the third network in the product is cumulative.



Self-citations $\mathbf{JCJ[SM, SM]}$

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Symbolic networks

Temporal bibliographic networks

Temporal citations between journals

Symbolic
networks

V. Batagelj

Introduction

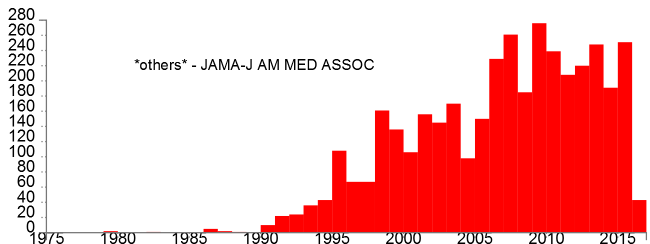
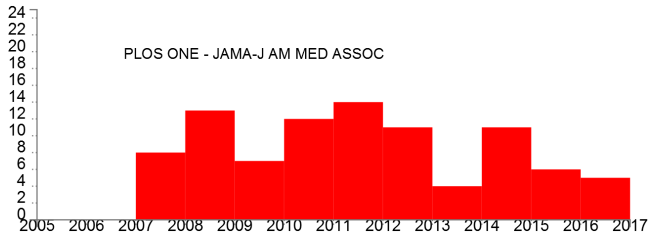
Networks and
semirings

Activity

Examples

Conclusions

References



Citations between journals $JCJ[PO, JAMA]$ and $inS(JCJ, JAMA)$



Conclusions

Symbolic
networks

V. Batagelj

Introduction

Networks and
semirings

Activity

Examples

Conclusions

References

Work in progress.

Another approach is based on event sequences. In applications, an event has additional properties besides the pair of nodes and starting time, such as duration, type, location, etc. [6].

Collections of interesting, understandable, and well-documented datasets (bikes, world trade, bibliographic data, political events (KEDS, WEIS), etc.). Sources of problems and ideas.

New analytical procedures: temporal clustering of nodes, temporal 1-neighbors, temporal cores, etc [7].

The computational work reported in this presentation was performed using **R**, Python, and Pajek. Some code and data are available at [GitHub/bavla/semirings](https://github.com/bavla/semirings).

This work was partially supported by the Slovenian Research Agency (research program P1-0294 and research project J5-4596) and was prepared within the framework of COST action CA21163 (HiTEc).

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